

Chapter 1: Sequences and Series of Real Numbers

A **sequence** of real numbers is an ordered list $(a_n)_{n=1}^{\infty}$. We say (a_n) *converges* to a limit $L \in \mathbb{R}$ if for every $\varepsilon > 0$ there exists N such that for all $n \geq N$, $|a_n - L| < \varepsilon$ ¹. In other words, the terms of the sequence eventually lie arbitrarily close to L . For example, $a_n = 1/n$ converges to 0 because eventually $1/n$ is within any given ε of 0 . By contrast, if no such L exists we say the sequence *diverges*.

A fundamental property of real numbers is **completeness**: every *Cauchy sequence* of real numbers converges. A sequence (x_n) is *Cauchy* if for every $\varepsilon > 0$ there is an index N such that for all $m, n \geq N$, $|x_m - x_n| < \varepsilon$ ². Equivalently, the terms eventually cluster arbitrarily tightly together. In $\mathbb{R}$ these two notions coincide: **Theorem (Completeness of $\mathbb{R}$)**: A sequence of real numbers is Cauchy if and only if it converges to a real limit ³. (The proof uses the Dedekind-completeness of $\mathbb{R}$ and can be found in standard analysis texts.)

An *infinite series* is the sum of the terms of a sequence. Formally, given (a_n) we write

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots,$$

and define the n -th *partial sum* $S_n = a_1 + a_2 + \dots + a_n$ ⁴. The series **converges** if the sequence (S_n) converges to a limit S . In that case we say $\sum a_n = S$. For instance, the geometric series $\sum_{n=0}^{\infty} r^n$ converges to $1/(1-r)$ when $|r| < 1$. A useful test for divergence is: **Divergence Test**: If $a_n \not\rightarrow 0$, then $\sum a_n$ diverges ⁵. Indeed, convergence of the series forces $a_n = S_n - S_{n-1} \rightarrow 0$. However, $a_n \rightarrow 0$ is only necessary, not sufficient (e.g. the harmonic series $\sum 1/n$ diverges).

Example: The sequence $a_n = (-1)^n$ does not converge (it oscillates). It is not Cauchy either, since terms do not get arbitrarily close. The geometric series $\sum_{n=0}^{\infty} 2^{-n}$ has partial sums $1 + 1/2 + \dots + 1/2^n = 2 - 2^{-n}$, which converge to 2 .

Exercises: 1. Prove that if (a_n) is increasing and bounded above, then it converges. 2. Show that the sequence $b_n = \sqrt{n^2 + 1} - n$ converges and find its limit. 3. Determine whether each series converges or diverges: (a) $\sum_{n=1}^{\infty} \frac{1}{n^p}$ for $p > 1$; (b) $\sum_{n=1}^{\infty} (-1)^n/n$; (c) $\sum_{n=1}^{\infty} 1/(n+(-1)^n)$. 4. Prove that every Cauchy sequence of real numbers is bounded. 5. Give an example of a Cauchy sequence that is not convergent in $\mathbb{Q}$ (rational numbers).

Chapter 2: Functions and Continuity

A **function** f from a set $D \subseteq \mathbb{R}$ to $\mathbb{R}$ assigns to each $x \in D$ a real value $f(x)$. We say f is *continuous* at a point $a \in D$ if

$$\lim_{x \rightarrow a} f(x) = f(a),$$

meaning for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - a| < \delta$ (with $x \in D$) implies $|f(x) - f(a)| < \varepsilon$ ⁶. Intuitively, $f(x)$ has no sudden jumps at a . The function $f(x) = x^2$ is continuous everywhere, while $g(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$ is discontinuous at 0 .

A function is *uniformly continuous* on D if the same δ works for all a in D ⁷. That is, $\forall \varepsilon > 0; \exists \delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$ for **all** $x, y \in D$. Uniform continuity is stronger; e.g. $f(x) = x^2$ on $[0, 1]$ is uniformly continuous, but on $\mathbb{R}$ it is not (since slopes grow without bound).

The Intermediate Value Theorem (IVT): If f is continuous on a closed interval $[a, b]$ and N is any number between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ with $f(c) = N$ ⁸. In simple terms, a continuous function on $[a, b]$ cannot skip over values; it takes on every intermediate value. For example, if $f(a) < 0 < f(b)$ then IVT guarantees a root $c \in (a, b)$ with $f(c) = 0$. The IVT follows from the completeness of $\mathbb{R}$ and can be proved by a bisection (or by considering the supremum of the set where $f(x) < N$).

The Extreme Value Theorem (EVT): If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum and minimum on $[a, b]$ ⁹. In other words, there exist $x_1, x_2 \in [a, b]$ such that $f(x_1) \leq f(x) \leq f(x_2)$ for all $x \in [a, b]$. For instance, a continuous temperature function over a day achieves a highest and lowest value. A common proof uses compactness: a continuous image of a compact set is compact, hence has max/min.

Example: The function $f(x) = x^3$ is continuous on $\mathbb{R}$, and on $[-1, 2]$ it achieves a minimum $f(-1) = -1$ and maximum $f(2) = 8$. By contrast, $g(x) = 1/x$ is not continuous at 0 , and on the open interval $(0, 1)$ it has no maximum (it blows up near 0).

Exercises:

1. Prove that polynomials and rational functions (where defined) are continuous using the ε - δ definition.
2. Show that any continuous function on a compact set is uniformly continuous.
3. Give an example of a function that is continuous everywhere but not uniformly continuous. (Hint: consider $f(x) = x^2$ on $\mathbb{R}$.)
4. Use the IVT to show that the equation $x^3 + x - 1 = 0$ has a root in $(0, 1)$.
5. Prove that if $f: [a, b] \rightarrow \mathbb{R}$ is continuous, then $f([a, b])$ is a closed and bounded interval.

Chapter 3: Sequences and Series of Functions

Consider a sequence of functions $\{f_n\}$ on a set A . We say f_n **converges pointwise** to f on A if for each fixed $x \in A$, $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$. In contrast, $\{f_n\}$ converges **uniformly** to f on A if

$$\lim_{n \rightarrow \infty} \sup_{x \in A} |f_n(x) - f(x)| = 0.$$

Equivalently, for every $\varepsilon > 0$ there exists N such that for all $n \geq N$ and all $x \in A$, $|f_n(x) - f(x)| < \varepsilon$ ¹⁰. Uniform convergence is a much stronger condition: it means eventually f_n is ε -close to f *everywhere* on A . Pointwise convergence can fail to preserve continuity, but uniform convergence often does.

Uniform Convergence Theorem: If each f_n is continuous on A and $f_n \rightarrow f$ uniformly on A , then f is continuous on A ¹¹. This theorem shows why uniform convergence is useful: it allows

swapping limit operations with continuity. Similarly, if each f_n is Riemann integrable on $[a, b]$ and $f_n \rightarrow f$ uniformly, then f is integrable and

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx$$

¹². In other words, one can interchange the limit and the integral when convergence is uniform. (In contrast, pointwise convergence alone does not guarantee these properties, as counterexamples show.)

For series of functions $\sum_{n=1}^{\infty} f_n(x)$, the **Weierstrass M-test** gives a simple criterion for uniform convergence: If there are constants $M_n \geq 0$ such that $|f_n(x)| \leq M_n$ for all $x \in A$ and $\sum_{n=1}^{\infty} M_n$ converges, then $\sum f_n(x)$ converges uniformly on A ¹³. Intuitively, each $f_n(x)$ is dominated by a term of a convergent numerical series. For example, the series $\sum_{n=1}^{\infty} \frac{\sin(nx)}{2^n}$ converges uniformly on all $\mathbb{R}$ because $|\sin(nx)| \leq 1$ and $\sum 2^{-n}$ converges.

Example: Let $f_n(x) = x/n$ on $[0, 1]$. Then $f_n(x) \rightarrow 0$ pointwise. Moreover, $|f_n(x)| \leq 1/n$ for all x , and $\sum 1/n$ diverges, so the M-test fails. Indeed f_n converges uniformly to 0 even though $\sum 1/n$ diverges (because $\sup_x |f_n(x) - 0| = 1/n \rightarrow 0$). In contrast, $g_n(x) = x^n$ on $[0, 1]$ converges pointwise to the function $g(x) = 0$ on $[0, 1)$ and $g(1) = 1$, but this convergence is not uniform (because near $x = 1$, x^n stays large).

Exercises:

1. Prove that if $\sum M_n$ converges and $|f_n(x)| \leq M_n$, then $\sum f_n(x)$ converges uniformly (Weierstrass M-test).
2. Give an example of a sequence of continuous functions $f_n: [0, 1] \rightarrow \mathbb{R}$ that converge pointwise but not uniformly.
3. Show by example that pointwise convergence does not in general allow exchanging limit and integral.
4. Verify that if $f_n(x) = \frac{1}{n} e^{-nx}$ on $[0, 1]$, then $f_n \rightarrow 0$ uniformly and $\int_0^1 f_n(x) dx \rightarrow 0$.
5. Let $f_n(x) = x^n$ on $[0, 1]$. Show that $f_n \rightarrow f$ pointwise, find $f(x)$, and determine whether convergence is uniform.

Chapter 4: Topology of the Real Line

The **open sets** of $\mathbb{R}$ provide a basic topology. A subset $E \subseteq \mathbb{R}$ is *open* if for every point $x \in E$ there exists an $\epsilon > 0$ such that the interval $(x - \epsilon, x + \epsilon)$ lies entirely in E ¹⁴. Equivalently, each point has a little “wiggle room” inside E . Standard examples: open intervals (a, b) are open, as is $\mathbb{R}$ itself (and also the empty set). Unions of open sets (even infinitely many) are open, and finite intersections of open sets are open.

A set F is *closed* if its complement $\mathbb{R} \setminus F$ is open. Equivalently, F contains all its limit points. In $\mathbb{R}$, one also shows (using sequences) that a set is closed iff it contains the limits of all convergent sequences drawn from it ¹⁵. For instance, $[a, b]$ is closed in $\mathbb{R}$, while $(a, b]$ is not.

A set $K \subseteq \mathbb{R}$ is **compact** if every open cover of K has a finite subcover. The Heine-Borel theorem characterizes compact subsets of $\mathbb{R}$: K is compact if and only if it is closed and bounded ¹⁶. In other words, closed intervals like $[a, b]$ are compact, whereas unbounded or non-

closed sets are not. Compactness has powerful consequences: any continuous function on a compact set is bounded and attains its extrema (used in the Extreme Value Theorem).

A set $S \subseteq \mathbb{R}$ is **connected** if it cannot be partitioned into two nonempty separated open sets. In $\mathbb{R}$ the following is true: **Proposition:** Every nonempty connected subset of $\mathbb{R}$ is an interval ¹⁷. Intuitively, in $\mathbb{R}$ no gap can occur in a connected set. Thus connected subsets are exactly the intervals (possibly infinite or singletons). For example, $[0,1] \cup [2,3]$ is disconnected because it splits into two separate pieces; $[0,1]$ is connected.

Exercises:

1. Prove that any closed interval $[a,b]$ is compact using the definition (cover by open intervals).
2. Show that $(0,1)$ is not compact by giving an explicit open cover without a finite subcover.
3. Verify that $\{1/n : n \in \mathbb{N}\}$ is a non-compact closed set (its limit point 0 is not included).
4. Prove that a continuous bijection from a compact set to $\mathbb{R}$ has a continuous inverse. (Hint: use the fact that continuous image of compact is compact.)
5. Show that any connected subset of $\mathbb{R}$ with more than one point must contain an interval. (That is, derive $\text{connected} \Rightarrow \text{interval}$.)

Chapter 5: The Riemann Integral

Let $f: [a,b] \rightarrow \mathbb{R}$ be a bounded function on a closed interval. A **partition** P of $[a,b]$ is a finite sequence $a = x_0 < x_1 < \dots < x_n = b$. On each subinterval $[x_{i-1}, x_i]$ define $m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}$ and $M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}$. The *lower Darboux sum* is

$$L(f, P) = \sum_{i=1}^n m_i (x_i - x_{i-1}),$$

and the *upper Darboux sum* is

$$U(f, P) = \sum_{i=1}^n M_i (x_i - x_{i-1}),$$

which approximate the “area” from below and above ¹⁸. As we refine the partition (making subintervals smaller), $L(f,P)$ increases and $U(f,P)$ decreases. We define the *lower integral* $L_f = \sup_P L(f,P)$ and *upper integral* $U_f = \inf_P U(f,P)$. If $L_f = U_f$, we say f is **(Riemann) integrable**, and the common value $\int_a^b f(x) dx = L_f = U_f$ ¹⁹.

Equivalently, f is integrable if for every $\epsilon > 0$ there exists a partition P such that $U(f,P) - L(f,P) < \epsilon$ ²⁰. This criterion is often used in proofs. A key fact is that if f is continuous on $[a,b]$ then it is integrable (continuous on compact $\Rightarrow$ integrable). More generally, bounded functions that are continuous except on a set of measure zero are integrable (this is Lebesgue’s criterion, see advanced texts).

For integrable functions, one proves linearity and additivity: if f, g are integrable then so are $f+g$ and cf , and $\int (f+g) = \int f + \int g$. Monotonicity holds: if $f(x) \leq g(x)$ then $\int_a^b f \leq \int_a^b g$. Moreover, one can use comparison tests: if $|f(x)| \leq M$ on $[a,b]$, then $|\int_a^b f| \leq M(b-a)$.

Improper Integrals: When the interval is unbounded or f has a discontinuity, we define integrals by limits. For example,

$$\int_a^\infty f(x) dx = \lim_{R \rightarrow \infty} \int_a^R f(x) dx,$$

if the limit exists ²¹. Similarly, if f has a vertical asymptote at b , we set

$$\int_a^b f(x) dx = \lim_{c \rightarrow b^-} \int_a^c f(x) dx.$$

These are called *improper Riemann integrals*. Convergence of such limits must be checked; e.g. $\int_1^\infty 1/x^p dx$ converges if and only if $p > 1$. Improper integrals extend the usual definite integral to many more functions, but some fail to converge (diverge).

Exercises:

1. Using Darboux sums, prove that a continuous function on $[a, b]$ is Riemann integrable.
2. Show that if f has finitely many jump discontinuities on $[a, b]$, then it is integrable. (Hint: around each discontinuity, exclude a small interval.)
3. Compute $\int_0^1 x^p dx$ for $p > -1$ by finding antiderivatives or using the definition.
4. Determine whether the improper integral $\int_1^\infty 1/(x(\ln x)^2) dx$ converges or diverges.
5. Establish the additivity of the integral: if $c \in [a, b]$, show $\int_a^b f = \int_a^c f + \int_c^b f$ for integrable f .

Chapter 6: Introduction to Probability Foundations

A **probability space** is a model $(\Omega, \mathcal{F}, P)$ where Ω is the sample space of outcomes, $\mathcal{F}$ is a collection of *events* (usually a σ -algebra of subsets of Ω), and $P: \mathcal{F} \rightarrow [0, 1]$ is a probability measure satisfying Kolmogorov's axioms ²² ²³:

- (Non-negativity) $P(E) \geq 0$ for every event $E \in \mathcal{F}$.
- (Normalization) $P(\Omega) = 1$.
- (Countable additivity) If $E_1, E_2, \dots$ are disjoint events, then $P(\bigcup_i E_i) = \sum_i P(E_i)$.

From these axioms follow the usual rules: $P(\emptyset) = 0$, $0 \leq P(E) \leq 1$, and $P(A^c) = 1 - P(A)$ for any event A . Probability theory studies functions (random variables) on this space and their distributions.

A **random variable** X is a function $X: \Omega \rightarrow \mathbb{R}$ (usually assumed measurable) that assigns a real value to each outcome. Intuitively, X encodes a numerical quantity of interest (e.g. number of heads in coin flips). Given X , one defines its *cumulative distribution function (CDF)* by

$$F_X(x) = P(X \leq x).$$

The CDF $F_X: \mathbb{R} \rightarrow [0, 1]$ is non-decreasing and right-continuous, with $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow +\infty} F_X(x) = 1$ ²⁴ ²⁵. It completely characterizes the distribution of X . For example, a *discrete* random variable takes countably many values $a_1, a_2, \dots$ with probabilities $p_i = P(X = a_i)$; its CDF jumps at each a_i by p_i . A *continuous* random variable has a continuous CDF; for an *absolutely continuous* case, one can define a density $f_X(x)$ with $F_X(x) = \int_{-\infty}^x f_X(t) dt$.

Examples: - A fair coin flip: $\Omega = \{\text{H}, \text{T}\}$, $P(\{\text{H}\}) = P(\{\text{T}\}) = 1/2$. Define $X(\omega) = 1$ if heads, 0 if tails. Then $P(X=1) = P(X=0) = 1/2$.

- Uniform continuous: If $X \sim U(a,b)$, then $P(X \in [c,d]) = (d-c)/(b-a)$ for $a \leq c < d \leq b$, and $F_X(x) = (x-a)/(b-a)$ for $x \in [a,b]$.
- Exponential or normal distributions: $X \sim \text{Exp}(\lambda)$ has $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. $X \sim N(\mu, \sigma^2)$ has CDF given by the Gaussian error function.

These basics set the stage for more advanced topics like expectation, convergence of random variables, and finally measure-theoretic probability (where one replaces the Riemann integral by Lebesgue integration).

Exercises:

1. List and prove the basic consequences of the probability axioms: e.g. $P(\emptyset) = 0$, $P(A^c) = 1 - P(A)$, and if $A \subseteq B$ then $P(A) \leq P(B)$.
2. Give the definition of a discrete vs continuous random variable and relate it to its CDF.
3. If X is a random variable with CDF $F(x)$, show that $P(a < X \leq b) = F(b) - F(a)$ for any $a < b$ ²⁶.
4. Describe the CDF of a Bernoulli random variable (takes values 0 or 1 with probabilities $1-p$ and p).
5. Suppose X, Y are discrete random variables with joint distribution $P(X=i, Y=j) = p_{ij}$. Define the marginal CDF of X and compute it in terms of the p_{ij} .

Sources: The material in this primer builds on standard real analysis texts and probability theory references ¹ ⁶ ⁸ ⁹ ¹³ ¹¹ ¹⁶ ²⁰ ²¹ ²⁷ ²⁵ ²², among others. These resources contain detailed proofs and additional discussion of all concepts presented.

¹ Math 413 – Convergence of Sequences of Real Numbers

https://connect.gonzaga.edu/asset/file/1514/ch02_1_convergence_v00.html

² Cauchy sequence - Wikipedia

https://en.wikipedia.org/wiki/Cauchy_sequence

³ 18.100A Fall 2020 Lecture 10: The Completeness of the Real Numbers and Basic Properties of Infinite Series

https://ocw.mit.edu/courses/18-100a-real-analysis-fall-2020/mit18_100af20_lec102.pdf

⁴ ⁵ 8.3: Series of Real Numbers - Mathematics LibreTexts

[https://math.libretexts.org/Under_Construction/Purgatory/Book%3A_Active_Calculus_\(Boelkins_et_al.\)/08%3A_Sequences_and_Series/8.03%3A_Series_of_Real_Numbers](https://math.libretexts.org/Under_Construction/Purgatory/Book%3A_Active_Calculus_(Boelkins_et_al.)/08%3A_Sequences_and_Series/8.03%3A_Series_of_Real_Numbers)

⁶ 6.1: An Analytic Definition of Continuity - Mathematics LibreTexts

[https://math.libretexts.org/Bookshelves/Analysis/Real_Analysis_\(Boman_and_Rogers\)/06%3A_Continuity_-_What_It_Isnt_and_What_It_Is/6.01%3A_An_Analytic_Definition_of_Continuity](https://math.libretexts.org/Bookshelves/Analysis/Real_Analysis_(Boman_and_Rogers)/06%3A_Continuity_-_What_It_Isnt_and_What_It_Is/6.01%3A_An_Analytic_Definition_of_Continuity)

⁷ 3.5: Uniform Continuity - Mathematics LibreTexts

[https://math.libretexts.org/Bookshelves/Analysis/Introduction_to_Mathematical_Analysis_I_\(Lafferriere_Lafferriere_and_Nguyen\)/03%3A_Limits_and_Continuity/3.05%3A_Uniform_Continuity](https://math.libretexts.org/Bookshelves/Analysis/Introduction_to_Mathematical_Analysis_I_(Lafferriere_Lafferriere_and_Nguyen)/03%3A_Limits_and_Continuity/3.05%3A_Uniform_Continuity)

⁸ Continuity and the Intermediate Value Theorem

<https://web.ma.utexas.edu/users/m408n/CurrentWeb/LM2-5-10.php>

⁹ Absolute and Local Extrema

<https://web.ma.utexas.edu/users/m408n/CurrentWeb/LM4-1-4.php>

¹⁰ ¹³ math.ucdavis.edu

https://www.math.ucdavis.edu/~hunter/m125a/intro_analysis_ch5.pdf

11 12 **RA Interchange of limits**

https://www.jirka.org/ra/html/sec_liminter.html

14 15 16 **math.umd.edu**

<https://math.umd.edu/~mboyle/courses/410/open.pdf>

17 **MathCS.org - Real Analysis: Proposition 5.3.3: Connected Sets in $\mathbb{R}$ are Intervals**

<https://mathcs.org/analysis/reals/topo/proofs/connint.html>

18 19 20 **Darboux integral - Wikipedia**

https://en.wikipedia.org/wiki/Darboux_integral

21 **Improper integral - Wikipedia**

https://en.wikipedia.org/wiki/Improper_integral

22 23 **Kolmogorov axioms of probability | The Book of Statistical Proofs**

<https://statproofbook.github.io/D/prob-ax.html>

24 25 26 **Cumulative distribution function - Wikipedia**

https://en.wikipedia.org/wiki/Cumulative_distribution_function

27 **Random variable - Wikipedia**

https://en.wikipedia.org/wiki/Random_variable